127. Which one of the following fruits is parthenocarpic?

(1) Jackfruit

(2) Banana

(3) Brinjal

(4) Apple

Balbiani rings are sites of :

(1) Polysaccharide synthesis

(2) RNA and protein synthesis

(3) Lipid synthesis

(4) Nucleotide synthesis

129. A pleiotropic gene :

controls a trait only in combination with another gene

controls multiple traits in an individual.

(3) is expressed only in primitive plants.

(4) is a gene evolved during Pliocene.

Grafted kidney may be rejected in a patient due to :

(1) Passive immune response

(2) Innate immune response

(3) Humoral immune response

(4) Cell-mediated immune response

131 During ecological succession :

(1) the numbers and types of animals remain constant.

(2) the changes lead to a community that is in near equilibrium with the environment and is called pioneer community

the gradual and predictable change in species composition occurs in a given area

(4) the establishment of a new biotic community is very fast in its primary phase.

132. Which of the following are **not** membrane - bound ?

(1) Lysosomes

(2) Mesosomes

(3) Vacuoles

(4) Ribosomes

133. Increase in concentration of the toxicant at successive trophic levels is known as :

(1) Biotransformation

(2) Biogeochemical cycling

(3) Biomagnification

(4) Biodeterioration

134. A jawless fish, which lays eggs in fresh water and whose ammocoetes larvae after metamorphosis return to the ocean is :

(1) *Neomyxine*

(2) *Petromyzon*

(3) *Eptatretus*

(4) *Myxine*

135. Choose the **wrong** statement :

(1) Morels and truffles are poisonous mushrooms

(2) Yeast is unicellular and useful in fermentation

(3) *Penicillium* is multicellular and produces antibiotics

(4) *Neurospora* is used in the study of biochemical genetics

136. Strong reducing behaviour of H_3PO_2 is due to

(1) High electron gain enthalpy of phosphorus

(2) High oxidation state of phosphorus

(3) Presence of two $-OH$ groups and one $P-H$ bond

(4) Presence of one $-OH$ group and two $P-H$ bonds

137. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence :

(1) Al < Ga < In < Tl

(2) ~~Tl < In < Ga < Al~~

(3) In < Tl < Ga < Al

(4) Ga < In < Al < Tl

138. The oxidation of benzene by V_2O_5 in the presence of air produces :

(1) maleic anhydride

(2) benzoic acid

(3) benzaldehyde

(4) ~~benzoic anhydride~~

139. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be :

(1) $\frac{1}{2}K$

(2) K

(3)

(4)

140. Gadolinium belongs to 4*f* series. It's atomic number is 64. Which of the following is the correct electronic configuration of gadolinium ?

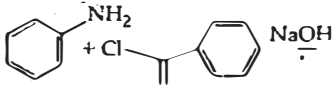
(1) [Xe] $4f^9 5s^1$

(2) [Xe] $4f^7 5d^1 6s^2$

(3) [Xe] $4f^6 5d^2 6s^2$

(4) [Xe] $4f^8 6d^2$

141. The following reaction



is known by the name

(1) Perkin's reaction

(2) Acetylation reaction

(3) Schotten-Baumen reaction

(4) Friedel-Craft's reaction

142. What is the mass of the precipitate formed when 50 mL of 16.9% so s mixed with 50 mL of 5.8% NaCl solution ?

(Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

(1) 3.5 g

(2)

(3) 14 g

(4) 28 g

143. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed ?

(1) 2.0

(2) 7.0

(3) 1.04

(4) 12.65

144. Method by which Aniline cannot be prepared

(1) degradation of benzamide with bromine in alkaline solution.

(2) reduction of nitrobenzene with H_2/Pd in ethanol.

(3 potassium salt of phthalimide treated wi chlorobenzene followed by hydrolysis wi[A] aqueous NaOH solution.

(4) hydrolysis of phenylisocyanide with acid solution

145. If Avogadro number N_A, is changed fro 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would ch

(1 the mass of one mole of carbon.

(2) the ratio of chemical species to each other in a balanced equation.

(3) the ratio of elements to each other in a compound.

(4) the definition of mass in units of grams.

146. The variation of the boiling points of the hydrogen halides is in the order HF > HI > HBr > HCl.

What explains the higher boiling point of hydrogen fluoride ?

(1) There is strong hydrogen bonding between HF molecules.

(2) The bond energy of HF molecules is greater than in other hydrogen halides.

(3) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule.

(4) The electronegativity of fluorine is much higher than for other elements in the group.

147. The number of structural isomers possible from the molecular formula C_3H_9N is :

(1) 5

(2) 2

(3) 3

(4) 4

148. Which of the statements given below is incorrect ?

(1) O_3 molecule is bent

(2) ONF is isoelectronic with O_2N^-

(3) OF_2 is an oxide of fluorine

(4) Cl_2O_7 is an anhydride of perchloric acid

O

The formation of the oxide ion, $O^{2-}(g)$, from oxygen atom requires first an exothermic and then an endothermic step as shown below :

$O(g) + e^- \rightarrow O^-(g)$; $\Delta_f H^\ominus = -141\ kJ\ mol^{-1}$

$O^-(g) + e^- \rightarrow O^{2-}(g)$; $\Delta_f H^\ominus = +780\ kJ\ mol^{-1}$

Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that,

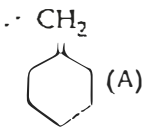
(1) O^- ion has comparatively smaller size than oxygen atom.

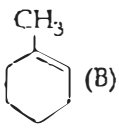
(2) oxygen is more electronegative

(3) addition of electron in oxygen results in larger size of the ion.

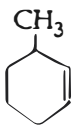
(4) electron repulsion outweighs the stability gained by achieving noble gas configuration.

1. In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is

(1)  (A)

(2)  (B)

(3) (A) and (B)

(4) 

1. 2,3 - Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid ?

(1) $(CH_3)_3C-CH=CH_2$

(2) $(CH_3)_2C=CH-CH_2-CH_3$

(3) $(CH_3)_2CH-CH_2-CH=CH_2$

(4) $(CH_3)_2CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH_2$

2. The hybridization involved in complex $[Ni(CN)_4]^{2-}$ is : (At. No. Ni = 28)

(1) sp^3

(2) d^2sp^2

(3) d^2sp^3

(4) dsp^2

153. Reaction of a carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is :

(1) hydrazine in presence of feebly acidic solution

(2) hydrocyanic acid

(3) sodium hydrogen sulphite

(4) a Grignard reagent

154. Two possible stereo-structures of $CH_3CHOH.COOH$, which are optically active, are called :

(1) Atropisomers

(2) Enantiomers

(3) Mesomers

(4) Diastereomers

155. In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with :

(1) carbon monoxide

(2) copper(I) sulphide

(3) sulphur dioxide

(4) iron(II) sulphide

156. A gas such as carbon monoxide would be most likely to obey the ideal gas law at :

(1) low temperatures and high pressures.

(2) high temperatures and high pressures.

(3) low temperatures and low pressures.

(4) high temperatures and low pressures.

157. In an S_N1 reaction on chiral centres, there is :

(1) inversion more than retention leading to partial racemization

(2) 100% retention

(3) 100% inversion

(4) 100% racemization

158. The vacant space in bcc lattice unit cell is :

(1) 48%

(2) 23%

(3) 32%

(4) 26%

159. The name of complex ion, $[Fe(CN)_6]^{3-}$ is :

(1) Hexacyanitoferrate (III) ion

(2) Tricyanoferrate (III) ion

(3) Hexacyanidoferrate (III) ion

(4) Hexacyanoiron (III) ion

160. The number of water molecules is maximum in :

(1) 1.8 gram of water
(2) 18 gram of water
(3) 18 moles of water
(4) 18 molecules of water

161. The heat of combustion of carbon to CO_2 is -393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is :

(1) +315 kJ
(2) −630 kJ
(3) −3.15 kJ
(4) −315 kJ

162. Aqueous solution of which of the following compounds is the best conductor of electric current ?

(1) Hydrochloric acid, HCl
(2) Ammonia, NH_3
(3) Fructose, $C_6H_{12}O_6$
(4) Acetic acid, $C_2H_4O_2$

163. In which of the following pairs, both the species are not isostructural ?

(1) diamond, silicon carbide
(2) NH_3, PH_3
(3) XeF_4, XeO_4
(4) $SiCl_4$, PCl_4^+

164. What is the mole fraction of the solute in a 1.00 m aqueous solution ?

(1) 1.770
(2) 0.0354
(3) 0.0177
(4) 0.177

165. Which of the following statements is not correct for a nucleophile ?

(1) Ammonia is a nucleophile
(2) Nucleophiles attack low e^- density sites
(3) Nucleophiles are not electron seeking
(4) Nucleophile is a Lewis acid

166. The sum of coordination number and oxidation number of the metal M in the complex $[M(en)_2(C_2O_4)]Cl$ (where *en* is ethylenediamine) is :

(1) 6
(2) 7
(3) 8
(4) 9

167. Assuming complete ionization, same mol which of the following compounds will requir *least* amount of acidified $KMnO_4$ for compl oxidation ?

(1) $FeSO_3$
(2) FeC_2O_4
$Fe(NO_2)_2$
(4) $FeSO_4$

168. Reaction of phenol with chloroform in presence dilute sodium hydroxide finally introduces whi one of the following functional group ?

(1) $-COOH$
(2) $-CHCl_2$
(2) $-CHO$
(3) $-CH_2Cl$

169. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?

(At. no. Z = 22)

(1) 4s 3s 3p 3d
(2) 3s 3p 3d 4s
(3) 3s 3p 4s 3d
(4) 3s 4s 3p 3d

170. Which one of the following pairs of solution is not an acidic buffer ?

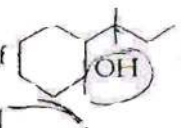
(1) CH_3COOH and CH_3COONa
(2) H_2CO_3 and Na_2CO_3
(3)
(4) $HClO_4$ and $NaClO_4$

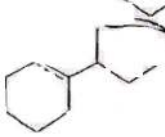
171. Number of possible isomers for the complex $[Co(en)_2Cl_2]Cl$ will be : (en = ethylenediamine)

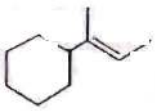
(1) 1
(2) 3
(3) 4
(4) 2

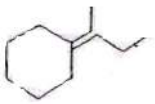
172. Decreasing order of stability of O_2, O_2^-, O_2^+ and O_2^{2-} is :

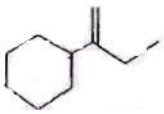
(1) $O_2^{2-} > O_2^- > O_2 > O_2^+$
(2) $O_2 > O_2^+ > O_2^{2-} > O_2^-$
(3) $O_2^- > O_2^{2-} > O_2^+ > O_2$
(4) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

173. Which of the following is not the product of dehydration of  ?

(1) 

(2) 

(3) 

(4) 

174. The correct statement regarding defects in crystalline solids is :

(1) Frenkel defects decrease the density of crystalline solids.

(2) Frenkel defect is a dislocation defect.

(3) Frenkel defect is found in halides of alkaline metals.

(4) Schottky defects have no effect on the of crystalline solids.

175. The rate constant of the reaction A → B is 3 mole per second. If the concentratio then concentration of B after 20 minutes

(1) 3.60 M

(2) 0.36 M

(3) 0.72 M

(4) 1.08 M

176. On heating which of the following releas most easily ?

(1) Na_2CO_3

(2) $MgCO_3$

(3) $CaCO_3$

(4) K_2CO_3

177. 20.0 g of a magnesium carbona e sample d on heating to give carbon dioxi magnesium oxide. What will be purity of magnesium carbonate ?

(1) 96

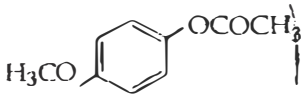
(2) 60

(3) 84

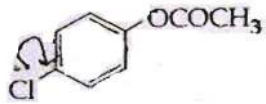
(4) 75

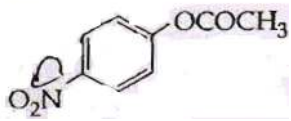
(At. Wt. : Mg = 24)

178. Which one of the following esters gets hydrolysed *most easily* under alkaline conditions ?

(1) 

(2) 

(3) 

(4) 

79. Caprolactam is used for the manufacture of :

(1) Teflon

(2) Terylene

(3) Nylon - 6, 6

(4) Nylon - 6

Which of the following reaction(s) can be used for the preparation of alkyl halides ?

(I) $CH_3CH_2OH + HCl \xrightarrow{\text{anh. } ZnCl_2}$

(II) $CH_3CH_2OH + HCl \longrightarrow$

(III) $(CH_3)_3COH + HCl \longrightarrow$

(IV) $(CH_3)_2CHOH + HCl$

(1) (I) and (II) only

(2) (IV) only

(3) (III) and (IV) only

(4) (I), (III) and (IV) only